Lab 6 - What Level of Agency?

ADS 5984 - Agentic AI in Practice

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Deliverable A

Table 1: Decision Matrix

Case	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Level	Why not higher?
1. Tempo detection and quantization	Y	N	N	N	N	N	Medium	Very low	No LLM	This case only requires a deterministic solution which takes a drummer's BPM and snap the hits to the nearest sixteenth-note grid. This can be done using code. Anything more complex will introduce the possibility of hallucinations and latency, which the case explicitly states that latency needs to be low as it will run in real time inside the audio engine. Also, LLMs cannot 'hear' per se.
2. Month-end ledger reconciliation	Y	N	N	N	N	N	Low	High	No LLM	This case only requires a deterministic solution which joins bank statements to a transaction and any ones that can't are flagged, something that can be done by code. Anything more complex will introduce the possibility of hallucinations and unnecessary latency, although latency is most likely not an issue in this case. The auditor also must be able to reproduce every match exactly which cannot be done when using an LLM.

Case	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Level	Why not higher?
3. Post-match social caption	N	Y	N	N	N	N	Medium	Low	Single-call LLM	This case now requires output that is language generated. So, using an LLM with the box score and other relevant features, we can have an LLM provide the one sentence response. No tool is required because the data is already fed into the prompt. There is also no multi-step workflow or agency required.
4. Support-inbox triage and reply	N	Y	Y	N	N	N	Medium	Low	Tool-augmented LLM	This case requires an LLM to triage emails based on its intent (language of email) and use customer order status from internal API call, this makes it a tool-augmented LLM. The solution is a single-pass, so no agentic solution. No workflow needed. Anything extra will increase latency and we have no need for agentic loops.
5. Quiz generator with citebacks	N	Y	Y	Y	N	N	Low	Medium	Determin-istic workflow	This case is a deterministic workflow as the developer is the one creating the multi-step RAG design. There is no need for an agent as the pipeline is the same every time and doesn't require loops or unpredictable next steps.

Case	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Level	Why not higher?
6. “A content-marketing team”	N	Y	Y	N	Y	N	Low	Medium	Single agent	This is a common case of a company thinking they need an agent for each different personality in their software. However, this design can be achieved with a single agent. There are no parallel blocks in this design or context overflow issues that would warrant the need for multiple agents. Anything more than one will four times the cost/latency.
7. Exploratory analysis of a new dataset	N	Y	Y	N	Y	N	High	Low	Single agent	This case requires agency because of the unpredictable nature of the problem this system is trying to solve. It doesn’t know what to complete next until it knows the previous result. It is sequential, so making it multi-agent would see no benefit at extra cost and latency.
8. Disaster-relief dispatch	N	Y	Y	N	Y	Y	Very Low	Very low	Multi-agent	Because of the independent nature of the depots, a multi-agent system allows communication between the depots to observe inventory and the road conditions. It’s also useful as decisions are made concurrently. With only having a single agent, the data ingestion may be too much and context will fill quickly. An agency will accept the higher usage bill if it means more lives are saved.

Case	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Level	Why not higher?
9. Systematic-review abstract triage	Y	N	N	N	N	N	Medium	Medium	No LLM	This case could solved using No LLM OR a single-call LLM depending on if the source material is labeled or not. If so, which I am assuming, an NLP model can be trained to classify the other documents. If no label, then an LLM would be a sufficient solution to this problem. There is no tool calling, workflow, or need for an agent, so a single LLM call will do.
10. "Radio-ready" mastering assistant	Y	N	N	N	N	N	Medium	Medium	No LLM	The solution for this case can be answered depending on what "radio-ready" means. I'm assuming there are various metrics the music must meet with hard numbers, so this could just be a simple DSP problem. LLMs could be used if there are verbal descriptions that must be met. Anything more than a single agent is probably too much, as a single agent can loop until it deems the music "radio-ready".

Case	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Level	Why not higher?
11. Plagiarism similarity check	N	Y	Y	N	N	N	Low	Medium	Workflow	The solution could utilize RAG where the prompt is the paper being checked and use ANN to find which chunks from the corpus of papers are similar to the ones written by the student. It would be a workflow with tool calls and the auditor can check with what the RAG responds with to determine if it's actual plagiarism (not quotes, names, etc.). An agentic solution would be excessive as any sort of loop would be unnecessary and would add latency, variance in answers, and context bloat.
12. Catalog alt-text generation	N	Y	N	N	N	N	Medium	Low	Single-call LLM	The solution only needs a single LLM call as no tools are needed, along with no workflows or agents - all of which would increase latency which is a real constraint for this case.
13. Make the failing test pass	N	Y	Y	N	Y	N	Low	Medium	Single agent	This case contains instances where each edit's results determines the next edit - meaning we need an agent to loop and reason until the tests have all passed. Multi-agent isn't needed as the process is all sequential and nothing can be done in parallel.

Case	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Level	Why not higher?
14. Live win-probability graphic	Y	N	N	N	N	N	Medium	Very low	No LLM	This case can be solved using a ML model trained on historical data. LLMs will make the results non-reproducible and without statistical grounding. Latency is also an issue where everything must be calculated live.
15. Help-center translation at scale	Y	N	N	N	N	N	Low	Low	No LLM	Instead of using an LLM, tools such as DeepL or Google translate exist which could translate the documents with little latency or cost, depending on the software. Anything more would be overkill and susceptible to hallucinations. Also, having 4,000 x 12 articles that need translating would probably run up the bill for a project that has cost at mind.
16. Emergency-department triage	N	Y	Y	Y	N	N	Very low	Low	Determin-istic workflow	This process is a deterministic workflow which can use an LLM to help take language from a patient and let already existing triaging methods determine the risk. This is important for auditing cases and prevents an LLM from making a wrong decision which could kill. There is a fixed number of steps and anything more will increase latency and consume life-changing time.

Case	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Level	Why not higher?
17. Meeting notes to action items	N	Y	Y	N	N	N	Medium	Medium	Tool-augmented LLM	This case requires LLMs to translate meeting transcript to actionable items and also need tools to write items to Calendar where necessary. It requires a single pass with no looping/branching - meaning no agents or workflows.
18. Red-team / blue-team hardening	N	Y	Y	N	Y	Y	Low	Medium	Multi-agent	This case requires Multi-agent as it requires a back-and-forth, reacting to each other's moves and needs to be context independent from each other. A single agent wouldn't grant this independence and anything less than a single agent wouldn't allow the continuous back-and-forth.
19. Adaptive next-exercise selection	Y	N	N	N	N	N	Low	Medium	No LLM	This approach doesn't even need an LLM because of the Item Response Theory mastery model that already exists. Feed results into model and have it output the necessary exercise. Although the case brings up a loop, the result of the loop is identical for every student, meaning no agent is needed as its predetermined. Anything more could result in hallucinations and cause the student to learn the wrong thing or something too easy or hard.

Case	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Level	Why not higher?
20. Clean up a messy CSV	Y	N	N	N	N	N	Low	Low	No LLM	If we assume that the data isn't too messy that the only cleaning that needs to be done are converting types, regexs, pivoting, etc., than a no LLM approach is adequate as all of these things can be done with code. The only reason why you might go up a level in complexity is if your cleaning rules are vague and the data is so messy that it would take a few loops to get it done.

Deliverable B

Table 2: Agency Level Comparison

Case	Level Before	Level After	Discriminating Question
1. Tempo detection and quantization	No LLM	No LLM	N/A
2. Month-end ledger reconciliation	No LLM	No LLM	N/A
3. Post-match social caption	Single-call LLM	Single-call LLM	N/A
4. Support-inbox triage and reply	Tool augmented LLM	Tool-augmented LLM	N/A
5. Quiz generator with citebacks	Deterministic workflow	Deterministic workflow	N/A
6. "A content-marketing team"	Multi-agent	Single agent	"Are there genuinely parallel sub-problems, or does one context overflow?"
7. Exploratory analysis of a new dataset	Single Agent	Single agent	N/A
8. Disaster-relief dispatch	Multi-agent	Multi-agent	N/A
9. Systematic-review abstract triage	Single agent	No LLM	"Does a deterministic / closed-form solution already exist?"
10. "Radio-ready" mastering assistant	No LLM	No LLM	N/A
11. Plagiarism similarity check	No LLM	Workflow	"Does a deterministic / closed-form solution already exist?"
12. Catalog alt-text generation	Single-call LLM	Single-call LLM	N/A
13. Make the failing test pass	Single agent	Single agent	N/A
14. Live win-probability graphic	No LLM	No LLM	N/A
15. Help-center translation at scale	No LLM	No LLM	N/A
16. Emergency-department triage	Single agent	Deterministic workflow	"Is the sequence of steps known and fixed in advance?"

Case	Level Before	Level After	Discriminating Question
17. Meeting notes to action items	Tool-augmented LLM	Tool-augmented LLM	N/A
18. Red-team / blue-team hardening	Multi-agent	Multi-agent	N/A
19. Adaptive next-exercise selection	No LLM	No LLM	N/A
20. Clean up a messy CSV	No LLM	No LLM	N/A

Deliverable C

Case 9

For this case, I am assuming that the research team already owns thousands of **labeled** documents. With this assumption, I concluded that the best architecture to solve this problem is a No LLM approach. Using the labeled data, one could train an NLP model and use it on the new documents that still need to be screened. My proposed level would change to a single-call LLM if my assumption was flipped - there wouldn't be any labeled documents to build a model off of, so we resort to an LLM.

Case 10

For this case, I am assuming that the definition of "radio-ready" follows some sort of fixed numeric target as opposed to verbal descriptions, or needing the system to listen, judge, and loop. I concluded that the best architecture to solve this problem is a No LLM approach where deterministic code can determine whether the music meets the requirements. If the assumptions were changed like the need for verbal descriptions, then something like a single-call LLM would be needed. Or, if we need to listen to the outputs and judge based on the goal, then use a single agent.

Case 20

For this case, I am assuming that the CSVs provided aren't too messy and that the rules are given up front such that there is no need for interpretation on column meanings, etc. I concluded that the best architecture to solve this problem with the assumption is a No LLM approach where deterministic code can read in the .csv, format it in pandas, and return the cleaned .csv. If the assumption that the data was less clean and no given rules, we could use a tool-augmented LLM where it can make interpretations on what needs to be done to make the dataframe clean without any given rules.

Deliverable D

In case 6, my first instinct was to use a multi-agent architecture for this problem, mostly due to the multiple number of personalities/roles that the solution required. However, there is no work that can be done in parallel - answering the question of "are there genuinely parallel sub-problems, or does one context overflow?". The overbuild would have skyrocketed the bill by 4x and would've have much larger latency without much change in reliability.